ESP32 Fleet Management — Tetradapt

Cloudflare + DigitalOcean — Single Droplet, All Open Source

1. Infrastructure Diagram — $24/mo

Internet | Cloudflare (Free) | DO Firewall (Free) | Droplet ($24) | Docker (Free) | Patient Home

INTERNET — Admins | Qualys | ESP32 OTA downloads

CLOUDFLARE (Existing Account — FREE)

DNS: fleet.tetradapt.com | SSL: Full (Strict) | CDN: Cache /firmware/* + /dashboard/* | DDoS + WAF

Orange Cloud Proxy | Rate Limit: /api/broadcast 10 req/min | Bypass Cache: /api/* | Edge TTL: firmware 1h, dashboard 2

DO Firewall FREE

443: CF IPs only

8883: Home IPs

51820: VPN

HTTPS :443

MQTTS :8883 (bypasses CF)

DO VPC — 10.0.0.0/16 — SFO3 (FREE)

SINGLE DROPLET — 2 vCPU / 4GB RAM / 80GB SSD — 4TB transfer $24/mo

DOCKER COMPOSE

- **Nginx** — :80 → CF handles SSL
- **FastAPI + Fleet API** — Fleet API + OTA + FCM Broadcast
- **React Dashboard** — Static → Nginx → CF CDN
- **Broadcast** — FCM Push
- **PostgreSQL 16** — Vol: /pgdata
- **Mosquitto MQTT** — :8883 TLS + ACL
- **WireGuard VPN** — :51820 UDP tunnels
- **Grafana OSS** — Dashboard :3000
- **Loki** — Log aggregation
- **Rsyslog** — UDP 514

Local Storage (80GB SSD)

/data/firmware/*.bin — served by Nginx, cached by Cloudflare CDN globally

Broadcast uses FCM push (no files stored) — firmware only

/data/backups/*.sql.gz — daily pg_dump, 7-day retention

/pgdata/ — PostgreSQL data volume

Backup: pg_dump daily → /data/backups/ (7-day retention) | DO Snapshots for DR ($0.06/GB)

DO Container Registry (Free: 1 repo, 500MB) | DO Monitoring (Free: CPU/RAM/Disk alerts)

WireGuard VPN (bypasses CF)

Home A
Community: NRH
Home Assistant + ESP32s + Sensors
WireGuard + MQTT | Fall sensor, Mobile App, Display

Home B
Community: Kaiser
Home Assistant + ESP32s + Sensors
WireGuard + MQTT | Motion, Door, Mobile App

Home N
Community: NRH + Sutter Health
Home Assistant + ESP32s + Sensors
WireGuard + MQTT | Sensors, Mobile App, Display

Total: $24/mo (just the Droplet) | Cloudflare: DNS + SSL + CDN + DDoS = FREE | Was $175/mo — saved 86%

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Application Components — Docker Compose on Single Droplet behind Cloudflare

2. Application Server Components

Cloudflare Edge | Nginx | FastAPI | MQTT | Fleet API | Auth | Background Svc | Dashboard

CLOUDFLARE EDGE — SSL Termination | CDN Cache (/firmware/*, /dashboard/*) | DDoS | WAF | Rate Limiting

SINGLE DROPLET — Docker Compose (2 vCPU / 4GB RAM) — $24/mo

NGINX :80 (SSL terminated at Cloudflare) — Reverse Proxy | Static Files | Firmware Serving | WebSocket

/api/* | /dashboard/* | /firmware/* | /ws/*

AUTH MIDDLEWARE (OAuth2/JWT) — admin: full access | operator: OTA + broadcast | viewer: read-only

FastAPI + Fleet API :8000

(No Celery/Redis — in-process scheduler)

REST API Routes

POST /api/broadcast → Community msg
GET /api/broadcast/:id → Delivery status
CRUD /api/communities | /homes
CRUD /api/devices | /firmware
POST /api/ota (rollouts)
GET /api/logs | /audit | /health

/ws/fleet-status | /ws/ota-progress

Internal Services

CommunityResolver group → home list
FirmwareValidator checksum + signing
AuditLogger | RateLimiter

Firmware Serving

/data/firmware/*.bin → Nginx → CF CDN
ESP32 downloads from CF edge (global)

React Dashboard (Nginx → CF CDN cached)

Fleet Overview | Device Detail | Log Viewer
OTA Manager | Communities | Security / Audit

Broadcast Center

Compose | Schedule | Community Select | Delivery Report

Real-time via WebSocket: fleet status, OTA progress

MQTT Client (aiomqtt)

Subscribes register, heartbeat, ack | Publishes (fleet topics only)

→ Mosquitto Docker (:8883 TLS)

Fleet API Tasks (in-process)

ota_rollout() | send_broadcast()

upload_firmware() | scheduled_broadcast() via FCM

Background Services

Heartbeat Mon. | Device Registrar | Alert Service

Scheduled FCM push | Dead device alerts → Email / Slack

ALL DOCKER CONTAINERS ON SAME DROPLET

PostgreSQL 16 | Mosquitto | WireGuard | Grafana | Loki | Rsyslog

Patient Homes via WireGuard VPN
NRH | Kaiser | St Jude | Sutter Health

All services: FREE (open source, self-hosted) — Only cost: $24/mo Droplet

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Data Flow — All traffic encrypted, Cloudflare edge caching, no PHI on server

3. Data Flow Paths

1. BROADCAST (FCM PUSH TO MOBILE APPS)

Admin → Dashboard → Cloudflare → POST /api/broadcast → Fleet API → FCM Push to subscribers → Home HA SmartHome App + Display → Patient hears

ACK → MQTT → Fleet API → PostgreSQL (delivery tracking) → Dashboard delivery report

2. OTA FIRMWARE UPDATE (Cloudflare CDN cached)

Admin → Upload .bin /api/firmware → Save → /data/ → Fleet API canary → staged → MQTT OTA cmd → ESP32 pulls .bin via CF CDN edge → Flash + Reboot

ACK → MQTT → PostgreSQL → rollback if failed

3. HEARTBEAT (every 30s via WireGuard — bypasses Cloudflare)

ESP32 Device → MQTT heartbeat (VPN) → Fleet API → store → PostgreSQL (Docker) → If dead >90s → Email / Slack / HA alert

4. DEVICE REGISTRATION (on boot via WireGuard)

ESP32 boots → MQTT register (MAC, ver) → Device Registrar → Auto-create in PostgreSQL → Assign to Home + Community

5. CENTRAL LOGGING

ESP32 syslog → UDP → Rsyslog (Docker) → Loki (index + search) → Grafana Dashboard (:3000)

6. DASHBOARD (real-time via Cloudflare)

Browser → CF Edge (SSL+CDN) → Nginx :80 → React (CF cached) + WSS upgrade → FastAPI WebSocket → fleet status | OTA progress

Traffic Encryption Summary

HTTPS: Browser → Cloudflare (SSL) → Origin Nginx (:80) | MQTTS: ESP32/HA → WireGuard VPN → Mosquitto :8883 TLS

WireGuard: UDP 51820 (Patient Homes ↔ Droplet, bypasses Cloudflare) | No PHI on fleet server or ESP32 — stays local in HA (HIPAA)